Algebraic Symmetry Breaking

Symmetry, Randomness, and Relational Structures

Michael L. Carroll

Key results of the monograph

Companion version 0.1.3 — November 2026

Roadmap

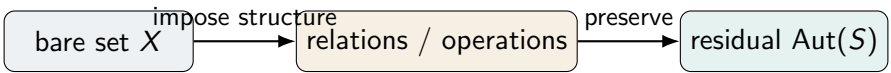
1. Thesis and scope
2. Finite relational structures
3. Relational ensembles
4. Mean field and critical behavior
5. Continuum and relational geometry
6. Algebra, infinity, and formalization
7. Reference-group phase diagnostics
8. Conclusions

The organizing question

Monograph source: Introduction

Can symmetry loss itself be the structural observable?

For a configuration or mathematical structure, measure not only a chosen order parameter, but the symmetry that survives the imposed relations.



Guiding principle

Structure is read as a reduction from available relabeling symmetry to residual automorphism symmetry.

One program, four levels of evidence

Monograph source: Introduction; Chapters 1–6; Appendices

1. **Exact finite theory:** automorphism groups, orbit sizes, refinement, and finite relational/algebraic ensembles.
2. **Exact model analysis:** a contrast-sensitive two-block free energy, its Hessian, and its fourth-order Landau jet.
3. **Numerical evidence:** mean-field, finite relational, Monte Carlo, and spatial lattice experiments.
4. **Prospective extensions:** Goldstone sectors, connections, curvature, renormalization, and infinite automorphism groups.

Claim discipline

The first two levels contain theorems. The third is empirical. The fourth supplies mathematically motivated frameworks and open problems, not a derivation of gauge theory or a universal infinite invariant.

Physical systems and three roles for infinity

Monograph source: Introduction; Chapter 5, Section 1; Chapter 6, Sections 1 and 5

- **Finite size, continuous variables:** N Ising spins have 2^N configurations; N classical rotors have configuration space $(S^1)^N$ and continuous global rotations.
- **Thermodynamic limit:** growing systems can develop sharp transitions; finite-state canonical systems with fixed finite energies have analytic free energies for $T > 0$.
- **Abstract infinite algebra:** a mathematical extension whose physical role depends on the model.

Physical reading path

Chapters 1–3 $\rightarrow$ Section 5.1 (signature, regularity, cardinal ratios) $\rightarrow$ Chapter 6. Chapter 4 and Sections 5.2–5.4 supply optional algebraic background as needed.

Public companion materials

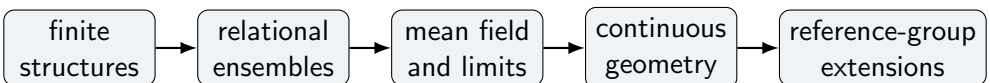
Monograph source: Companion Materials guide; Appendices A and D

`https://github.com/musicarroll/symmetry-companion`

- **Python:** models in `src/`, scans and plots in `scripts/`; start with `README_computational.md`.
- **Lean:** modules in `SymmetryMonograph/`; start with `lean/README.md` and the statement coverage report.
- **Recorded results:** data and figures in `results/`.
- **Reproducibility:** record the Git commit, software versions, parameters, and seeds. Proof scopes remain explicit in the ledger.

Narrative arc

Monograph source: Chapters 1–6



Invariant carried forward

available symmetry / preserved symmetry.

What changes

Counting → entropy density → dimension, measure, or asymptotic rate.

Roadmap

1. Thesis and scope
2. Finite relational structures
3. Relational ensembles
4. Mean field and critical behavior
5. Continuum and relational geometry
6. Algebra, infinity, and formalization
7. Reference-group phase diagnostics
8. Conclusions

Structures and automorphisms

Monograph source: Chapter 1, Section 1

A finite relational structure is

$$S = (X, \mathcal{R}), \qquad |X| = n,$$

and $f : X \to X$ is an automorphism when it preserves every relation:

$$(x_1, \ldots, x_k) \in R \iff (f(x_1), \ldots, f(x_k)) \in R.$$

No imposed relations

$\mathrm{Aut}(X) \cong S_n$: every relabeling survives.

Rigid structure

$\mathrm{Aut}(S) = \{e\}$: the relations distinguish every point.

$\mathrm{Aut}(S) \leq \mathrm{Sym}(X)$ turns "structure" into a subgroup-comparison problem.

The finite symmetry deficit

Monograph source: Chapter 1, Section 2

Definition

For S on an n-element set,

$$I(S) = \log\left(\frac{n!}{|\operatorname{Aut}(S)|}\right).$$

The full symmetric group acts on labeled presentations of S, with stabilizer $\operatorname{Aut}(S)$. Hence orbit–stabilizer gives

$$|\operatorname{Orb}(S)| = \frac{|S_n|}{|\operatorname{Aut}(S)|} = e^{I(S)}.$$

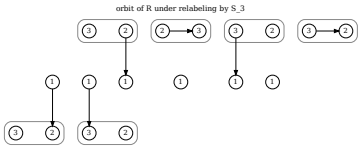
Two exact meanings of the same number

$I(S)$ is both logarithmic symmetry loss and the logarithm of the number of label-distinct realizations of the isomorphism class.

Small examples calibrate the invariant

Monograph source: Chapter 1, Section 2

structure on X	$\lvert \mathrm{Aut}(S) \rvert$	$I(S)$	interpretation
no relations, $\lvert X \rvert = n$	$n!$	0	fully symmetric
one distinguished point	$(n-1)!$	$\log n$	one orbit choice
one loop on 3 points	2	$\log 3$	two points remain exchangeable
one arrow on 3 points	1	$\log 6$	rigid



The six relabelings form one S_3-orbit because the stabilizer of a single directed arrow is trivial.

For every finite relational structure S on n points:

$$0 \le I(S) \le \log(n!),$$
$$I(S) = 0 \iff \mathrm{Aut}(S) = S_n,$$
$$I(S) = \log(n!) \iff S \text{ is rigid.}$$

Theorem (Monotonicity under refinement)

If S_2 retains every relation symbol of S_1 and adds further relational data, then

$$\mathrm{Aut}(S_2) \le \mathrm{Aut}(S_1) \implies I(S_2) \ge I(S_1).$$

Proof idea: every automorphism preserving the richer signature also preserves its reduct; take subgroup orders and logarithms.

A crucial non-monotonicity

Monograph source: Chapter 1, Section 4

Adding named relations

Passing from (X, R) to (X, R, Q) retains information and can only shrink the automorphism group.

Combining tuple sets

Replacing R, Q by $R \cup Q$ or $R \cap Q$ may forget which tuple came from which relation. Symmetry deficit may rise or fall.

Exact invariances over a fixed $X \times X$ include

$$I(R) = I(R^{c}) = I(R^{-1}).$$

Lesson: monotonicity is a theorem about refinement of structure, not about set inclusion between tuple sets.

Roadmap

1. Thesis and scope
2. Finite relational structures
3. Relational ensembles
4. Mean field and critical behavior
5. Continuum and relational geometry
6. Algebra, infinity, and formalization
7. Reference-group phase diagnostics
8. Conclusions

The relational partition function

Monograph source: Chapter 2, Section 1

Let $E(S)$ be isomorphism-invariant. Summing over isomorphism classes,

$$Z(\beta) = \sum_{[S]} \frac{n!}{|\operatorname{Aut}(S)|} e^{-\beta E(S)} = \sum_{[S]} e^{I(S) - \beta E(S)}.$$

The induced Gibbs law is

$$\mathbb{P}_\beta([S]) = \frac{e^{I(S) - \beta E(S)}}{Z(\beta)}.$$

Interpretation

Energy weights structures; each orbit counts labeled realizations. X labels entities; configurations are structures on X.

Physical use requires distinct admissible configurations. Identify redundant labels first and specify the background-preserving reference action.

Energy–multiplicity competition

Monograph source: Chapter 2, Sections 1, 2, and 5

Highly symmetric type

- large $|\operatorname{Aut}(S)|$,
- small $I(S)$,
- few labeled realizations.

Rigid type

- $|\operatorname{Aut}(S)| = 1$,
- $I(S) = \log(n!)$,
- a full orbit of $n!$ realizations.

For singleton relations on three points, a rewarded loop contributes $3e^{\beta\varepsilon}$ while a directed edge contributes 6. Thus the loop wins exactly when $\beta\varepsilon > \log 2$.

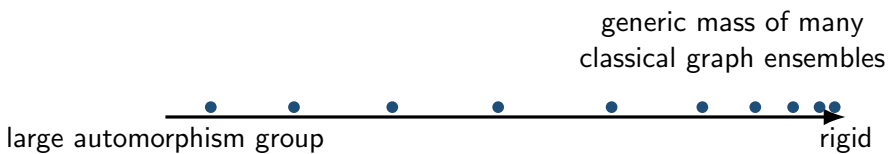
Not thermodynamic entropy by definition

$I(S)$ is an orbit-counting entropy-like term. Its physical meaning depends on the ensemble and on which labels represent genuine microstates.

Why asymmetry is statistically relevant

Monograph source: Chapter 2, Section 2

In the classical uniform ensemble of undirected graphs, Erdős and Rényi proved that almost all large graphs are asymmetric [1].



Scope of the precedent

It supports the importance of rigid orbit types, but it is not a theorem about every relational ensemble, every energy, or law-constrained algebraic structures.

Roadmap

1. Thesis and scope
2. Finite relational structures
3. Relational ensembles
4. Mean field and critical behavior
5. Continuum and relational geometry
6. Algebra, infinity, and formalization
7. Reference-group phase diagnostics
8. Conclusions

Coarse graining by block densities

Monograph source: Chapter 3, Sections 1–2

Given an ordered partition

$$\mathcal{P} = (X_1, \dots, X_k), \qquad X = X_1 \sqcup \cdots \sqcup X_k,$$

the *block profile* is the coarse tuple

$$\mathrm{prof}_{\mathcal{P}}(S) = (\alpha, \rho), \quad \alpha_i = \frac{|X_i|}{|X|}, \quad \rho_{ij} = \frac{|R \cap (X_i \times X_j)|}{|X_i||X_j|}.$$

For fixed counts $m_{ij} = |R \cap (X_i \times X_j)|$, the labeled degeneracy is

$$\prod_{i,j} \binom{|X_i||X_j|}{m_{ij}}.$$

Using Stirling's approximation with $h(u) = -u \log u - (1-u)\log(1-u)$ gives the entropy density

$$\mathcal{S}_k(\alpha, \rho) = \sum_{i,j} \alpha_i \alpha_j h(\rho_{ij}).$$

Variational free energy

Monograph source: Chapter 3, Sections 1–3

If $E(S) \approx n^2 \Phi_k(\alpha, \rho)$, then profiles contribute at leading exponential order as

$$Z(\beta) \approx \sum_{\text{profiles}} \exp\left[-n^2 \mathcal{F}_\beta(\alpha, \rho)\right],$$

where

$$\boxed{\mathcal{F}_\beta = \beta \Phi_k(\alpha, \rho) - \sum_{i,j} \alpha_i \alpha_j h(\rho_{ij}).}$$

Homogeneous profile

All ρ_{ij} coincide; block labels are macroscopically invisible.

Differentiated profile

Within- and cross-block densities differ; the partition becomes structurally visible.

Coarse degeneracy groups many exact isomorphism classes; it is an asymptotic replacement for, not an identity with, exact orbit multiplicity.

The contrast-sensitive two-block model

Monograph source: Chapter 3, Section 4

For equal blocks, impose

$$\rho_{11} = \rho_{22} = p, \qquad \rho_{12} = \rho_{21} = q.$$

The refined free energy is

$$\boxed{F_\beta(p,q) = -\frac{\beta J}{2}(p-q)^2 - h(p) - h(q), \qquad 0 < p, q < 1.}$$

It is invariant under block exchange $(p,q) \mapsto (q,p)$. Define

$$m = \frac{p+q}{2}, \qquad d = \frac{p-q}{2}, \qquad \Delta = |p-q| = 2|d|.$$

Model logic

Entropy favors $p = q = 1/2$; the energy rewards contrast. The antisymmetric coordinate d isolates block differentiation.

Exact Hessian instability

Monograph source: Chapter 3, Section 4

At the symmetric point $(p, q) = (1/2, 1/2)$,

$$H = \begin{pmatrix} 4 - \beta J & \beta J \\ \beta J & 4 - \beta J \end{pmatrix}.$$

mode	eigenvector	eigenvalue
mean-density	$(1, 1)$	$\lambda_+ = 4$
block-contrast	$(1, -1)$	$\lambda_- = 4 - 2\beta J$

$\beta_c = \dfrac{2}{J}$ is the exact loss-of-local-stability threshold in the antisymmetric mode.

Logical boundary

A negative Hessian direction proves instability, not by itself the existence of a nearby nonzero stationary branch.

The fourth-order Landau jet

Monograph source: Chapter 3, Section 5

Restrict to the symmetric contrast slice $m = 1/2$:

$$\mathcal{G}_{\beta,J}(d) = -2\beta J d^2 - h(1/2 + d) - h(1/2 - d).$$

Its exact derivatives through order four give

$$J_0^4 \mathcal{G}(d) = F_0 + (4 - 2\beta J)d^2 + \frac{8}{3}d^4.$$

For the quartic approximation and $\beta > \beta_c$,

$$d_*^2 = \frac{3}{8}(\beta J - 2), \qquad \Delta \sim (\beta - \beta_c)^{1/2}, \qquad D_{\mathrm{cont}} = (p - q)^2 \sim (\beta - \beta_c).$$

Formal status

Lean verifies the jet and, for $|d| \le r < 1/2$, the uniform bound

$$|\mathcal{G}(d) - J_0^4 \mathcal{G}(d)| \le \frac{128}{1 - 4r^2}|d|^6.$$

The fixed-mean expansion also has a Lean proof of its $O(d^6)$ remainder for every $0 < m < 1$.

The exact broken minimum pair is unique

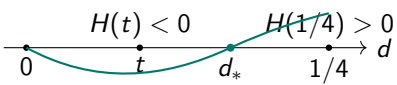
Monograph source: Chapter 3, Section 5

Let $H_{\beta,J}(d) = \mathcal{G}'_{\beta,J}(d)$. On $|d| < 1/2$,

$$H_{\beta,J}(d) = 0 \quad \Longleftrightarrow \quad \nabla F_\beta(1/2 + d, 1/2 - d) = 0.$$

If $2 < \beta J < 2\log 3$, then

$$H'(0) = 8 - 4\beta J < 0, \qquad H(1/4) = -\beta J + 2\log 3 > 0.$$



Continuity gives $d_* \in (0, 1/4)$; oddness gives $-d_*$. Strict monotonicity of the exact coupling map makes this pair unique, and entropy concavity plus positive exact slice curvature makes both points **full local minima**.

The exact branch has the Landau exponent

Monograph source: Chapter 3, Section 5

Parameterize the positive exact branch by

$$K = \beta J = \kappa(d) := \frac{1}{2d} \log\left(\frac{1+2d}{1-2d}\right), \qquad 0 < d < \frac{1}{2}.$$

The derivative remainder gives the explicit normalized estimate

$$\left| \frac{\kappa(d) - 2}{d^2} - \frac{8}{3} \right| \leq \frac{32d^2}{1-4d^2}.$$

Therefore, as $d \downarrow 0$ (equivalently $K \downarrow 2$),

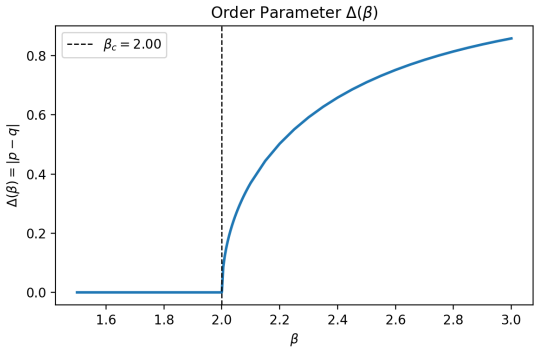
$$\frac{d^2}{K-2} \longrightarrow \frac{3}{8}, \qquad d \sim \sqrt{\frac{3}{8}(K-2)}.$$

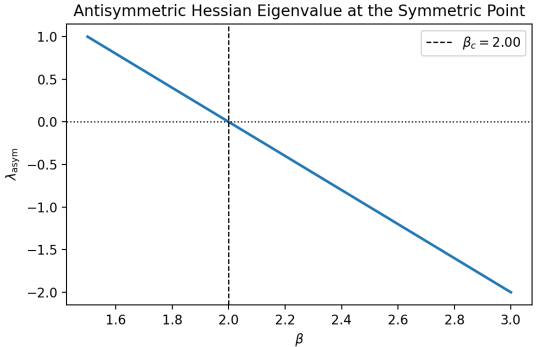
Exact conclusion

The square-root law is now a theorem for the unique exact branch, not only the prediction of the fourth-order Landau polynomial.

Numerical onset matches the analytic threshold

Monograph source: Chapter 3, Section 11; Appendix B

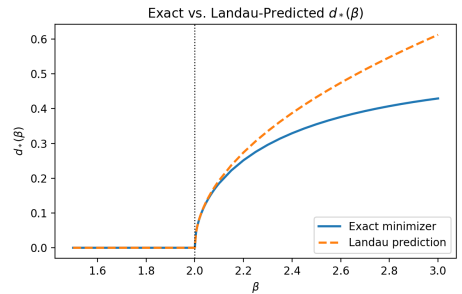




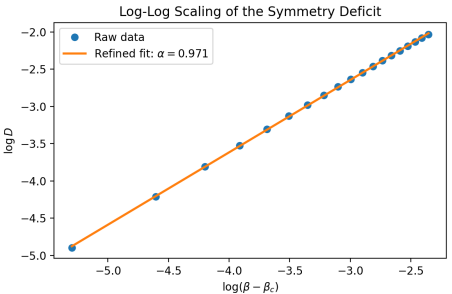
- $J = 1$ predicts $\beta_c = 2$.
- On the refined 0.005 grid, the first nonzero contrast occurs at 2.005.
- At $\beta = 2.025$, $(p_*, q_*) = (0.595760, 0.404240)$ and $\Delta = 0.191520$.

Landau scaling is local, not global

Monograph source: Chapter 3, Section 5; Appendix B



numerical branch versus quartic prediction



Near-threshold fit: slope $\alpha \approx 0.971$

Exact

Reading the evidence

The branch and $D_{\mathrm{cont}} \propto \beta - \beta_c$ scaling agree near onset. Deviations away from the critical neighborhood are expected because the quartic jet discards higher-order terms.

Residual symmetry versus contrast magnitude

Monograph source: Chapter 3, Section 11

Let $\Gamma = \{e, \tau\} \cong S_2$ act by $\tau(p, q) = (q, p)$. Define

$$I_\Gamma(p, q) = \log \frac{|\Gamma|}{|\operatorname{Stab}_\Gamma(p, q)|}.$$

Then exactly

$$I_\Gamma(p, q) = \begin{cases} 0, & p = q, \\ \log 2, & p \neq q. \end{cases} \qquad I_\Gamma = 0 \iff \Delta = 0.$$

Discrete invariant I_Γ

Detects parameter-sector exchange, not block relabeling.

Continuous proxy D_{cont}

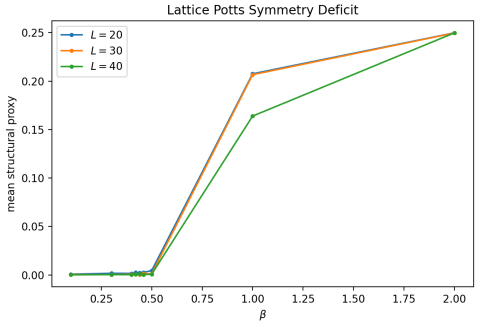
$(p - q)^2$ measures how far the state lies from the fixed locus.

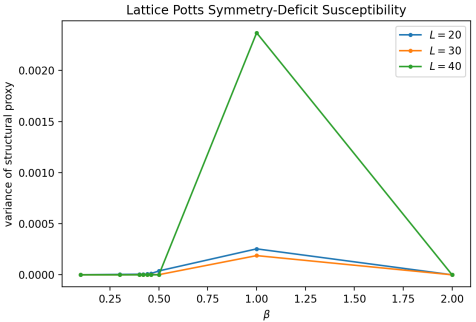
Spatial test: a two-state lattice model

Monograph source: Appendices B–C

For Potts spins $s_x \in \{1, \dots, q\}$ on an $L \times L$ periodic lattice,

$$E = -J \sum_{\langle x,y \rangle} \delta(s_x, s_y), \qquad D_{\mathrm{lat}} = \frac{1}{q} \sum_{i=1}^{q} \left(f_i - \frac{1}{q} \right)^2 .$$





For $L = 20, 30, 40$, the structural observable detects ordering and the sparse grid does not resolve critical scaling ($\beta_c J \simeq 0.881374$).

What the lattice data do—and do not—show

Monograph source: Appendix C

Supported

- D_{lat} is small in the symmetric regime.
- It approaches the ordered two-state value at large β.
- Saved peak widths depend on a sparse grid and short runs.

Not yet supported

- stable pseudocritical points,
- reliable critical exponents,
- universality-class identification,
- superiority to standard observables.

Current raw-variance fits have poor quality and are not estimates of γ/ν. If $D \sim m^2$, the structural hypothesis predicts $L^d \operatorname{Var}(D) \sim L^{d-4\beta_{\mathrm{mag}}/\nu}$, giving $3/2$ rather than $7/4$ in the 2D Ising class. A joint scaling collapse is required.

Roadmap

1. Thesis and scope
2. Finite relational structures
3. Relational ensembles
4. Mean field and critical behavior
5. Continuum and relational geometry
6. Algebra, infinity, and formalization
7. Reference-group phase diagnostics
8. Conclusions

From blocks to kernels

Monograph source: Chapter 3, Section 2

A k-block density matrix determines a step function

$$\rho_k : [0,1]^2 \to [0,1].$$

As the partition is refined, this motivates a measurable continuum profile and a functional

$$\mathcal{F}_\beta[\rho] = \beta\Phi[\rho] - \int_0^1 \int_0^1 h(\rho(x,y))\, dx\, dy.$$

Definition

An undirected graphon is a symmetric measurable $W : [0,1]^2 \to [0,1]$ [3].

Directed relational kernels need not satisfy $W(x,y) = W(y,x)$. Dense graph limits therefore supply a rigorous neighboring theory, but not every kernel model is an undirected graphon model.

Continuous symmetry breaking

Monograph source: Chapter 3, Sections 6–7

Consider a differentiable $O(m)$-invariant order-parameter field, $m \geq 1$,

$$\mathcal{F}[\phi] = \int_\Omega \left(\frac{\kappa}{2} |\nabla \phi|^2 + a|\phi|^2 + b|\phi|^4 \right) dx, \qquad b, \kappa > 0.$$

For $a < 0$, constant minima satisfy

$$|\phi| = v = \sqrt{\frac{-a}{2b}}, \qquad O(m) \longrightarrow O(m-1),$$

and the vacuum manifold is

$$O(m)/O(m-1) \cong S^{m-1}.$$

In the constrained vacuum model with $m = 2$, writing $\phi = v(\cos\theta, \sin\theta)$ gives

$$\Delta\mathcal{F} = \frac{\kappa v^2}{2} \int_\Omega |\nabla\theta|^2 dx.$$

Uniform rotations are free, and long-wavelength twists are arbitrarily soft.
Expanding about one vacuum gives one massive radial fluctuation and $m-1$ transverse directions with no quadratic potential term—the variational analogue of Goldstone modes [2].

Connections encode relational comparison

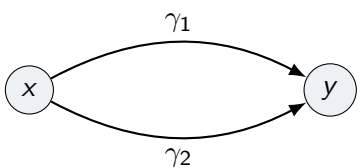
Monograph source: Chapter 3, Sections 8–9

A manifold-valued field does not by itself require a gauge field. As additional bundle/frame data, introduce

$$A \in \Omega^1(\Omega, \mathfrak{g}), \qquad D = d + A,$$

as an infinitesimal rule for comparing local orientations.
Its curvature is

$$F = dA + A \wedge A, \qquad [D_i, D_j]\phi = F_{ij} \cdot \phi.$$



infinitesimal path discrepancy
detected by curvature

Curvature dynamics

Monograph source: Chapter 3, Section 10

Given a positive-definite invariant inner product on $\mathfrak{g}$ and a background Hodge star, define

$$S[A] = \int_\Omega \mathrm{Tr}(F \wedge *F).$$

Compactly supported smooth variations of a smooth connection yield

$$\delta F = D(\delta A), \qquad \boxed{D * F = 0.}$$

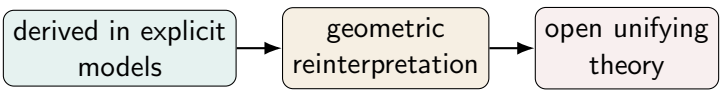
For $G = U(1)$ this gives $d * F = 0$ (with $dF = 0$); spacetime dynamics requires a Lorentzian metric; for nonabelian G it has Yang–Mills form [6].

Interpretive status

Standard connection–curvature mathematics is being given a relational reading. The book does not derive Yang–Mills theory from the finite formula $I(S) = \log(n!/|\operatorname{Aut}(S)|)$.

Where the continuum program stands

Monograph source: Chapter 3, Sections 6–12



Block variational calculus; Hessian; Landau jet; $O(m)$ model.

Local orientations as G/H fields; connections as comparison; curvature as obstruction.

Derive continuum geometry from microscopic ensembles; construct an RG flow; relate I to Fisher–Rao geometry.

Roadmap

1 Thesis and scope

2 Finite relational structures

3 Relational ensembles

4 Mean field and critical behavior

5 Continuum and relational geometry

6 Algebra, infinity, and formalization

7 Reference-group phase diagnostics

8 Conclusions

Operations are constrained relations

Monograph source: Chapter 4, Sections 1–3

A binary operation $\mu : X \times X \to X$ is encoded by

$$\Gamma_\mu = \{(x, y, z) \in X^3 : \mu(x, y) = z\}.$$

The relation is functional and total in the third coordinate:

$$\forall x, y \in X \; \exists! z \in X : \; (x, y, z) \in \Gamma_\mu.$$

Theorem

A permutation $f : X \to X$ preserves μ iff it preserves Γ_μ:

$$f(\mu(x, y)) = \mu(f(x), f(y)) \iff f^{\times 3}(\Gamma_\mu) = \Gamma_\mu.$$

Thus finite groups, rings, semigroups, lattices, and general universal algebras enter the same automorphism framework.

A complete two-element operation ensemble

Monograph source: Chapter 4, Section 4

There are n^{n^2} labeled binary operations on an n-element set; for $n = 2$, this is 16.

$\lvert \mathrm{Aut} \rvert$	classes	labeled tables
1	6	12
2	4	4
total	10	16

constraint	labeled	classes
associative	8	5
commutative	8	4
monoid	4	2
group	2	1

Algebraic laws define sharply smaller subensembles, while orbit multiplicity continues to be $n!/\lvert \mathrm{Aut}(A) \rvert$.

In infinity, the signature becomes decisive

Monograph source: Chapter 5, Sections 1–2

underlying set	specified structure	residual symmetry
$\mathbb{Z}$	additive group	$n \mapsto \pm n$
$\mathbb{Z}$	unital ring	identity only
$\mathbb{Z}$	ordered unital ring	identity only
$\mathbb{R}$	$\mathbb{Q}$-vector space	many linear automorphisms
$\mathbb{R}$	field	identity only
$\mathbb{R}^n$	vector space	$\mathrm{GL}_n(\mathbb{R})$
$\mathbb{R}^n$	inner-product space	$O(n)$

Infinite groups of the same cardinality may represent radically different symmetry. Consequently,

$$\log \frac{|\operatorname{Aut}(X)|}{|\operatorname{Aut}(A)|}$$

is generally not a meaningful numerical invariant.

Candidate infinite symmetry deficits

Monograph source: Chapter 5, Sections 3–4; Chapter 6

candidate	natural setting	principal limitation
$\log[G : H]$	finite-index residual symmetry	often infinite
$\dim G - \dim H$	Lie/algebraic groups	misses wild groups
quotient measure	compact/measured groups	needs invariant measure
finite-window rate	growing physical domains	boundary dependent
measure relative entropy	phase laws on G/H	needs reference measure

Reference-relative requirement

Choose a structured background B and $G \leq \mathrm{Aut}(B)$. Compare the phase stabilizer H with G; $G = \mathrm{Sym}(X)$ is reserved for a genuinely bare set.

Relational scaling taxonomy

Monograph source: Chapter 6, Section 6

quantity	question answered	caution
$I_G = \log[G : H]$	which finite sector is selected?	may be subextensive
$\widetilde{I}_G = I_G / \log \lvert G \rvert$	what fraction of G is lost?	not a density
I_L / a_L	is symmetry loss extensive?	declare a_L
$D(\rho), \widehat{D}$	is a coarse pattern visible?	model-dependent proxy
$a_L \operatorname{Var}(Q_L)$	how do fluctuations scale?	depends on Q_L

Reporting rule

Use total deficit for discrete selection and a declared deficit rate for extensive ordering. Never identify a block proxy with a normalized automorphism deficit without an explicit relation between them.

Machine-checked core

Monograph source: Appendix D; Lean sources and formalization ledger

Every numbered statement is paired with a checked ledger entry.

numbered-statement status	count
complete	46
partial	1
needs definitions	1
analysis-heavy	7
external citation	1
compatibility condition	1
total	57

- Chapters 1–2 and 4 connect automorphism subgroups, relabeling orbits, finite partition sums, and extremal equality cases.
- Chapter 3 checks general-mean $O(d^6)$ control, quartic local minima, the inverse-branch $3/8$ limit, and orthogonal field-energy invariance.
- Chapter 6 also checks Haar index, relative entropy, and scalar-unitary invariance of fixed-number and one-particle operators.
- Continuum limits, quotient geometry, gauge variation, and XY analysis remain open; the condensation theorem is an external citation.

Appendix D records the proof scope. Two unnumbered identities and nine supplementary targets are tracked separately.

Roadmap

1. Thesis and scope
2. Finite relational structures
3. Relational ensembles
4. Mean field and critical behavior
5. Continuum and relational geometry
6. Algebra, infinity, and formalization
7. Reference-group phase diagnostics
8. Conclusions

Physical symmetry is relative to a background

Monograph source: Chapter 6, Section 1

A physical domain carries background structure B before a phase is chosen: geometry, adjacency, species, interactions, locality, and boundary data.

$$\boxed{G \leq \mathrm{Aut}(B)}, \qquad H_S = \{g \in G : g \cdot S = S\}.$$

For finite index,

$$I_G(S) = \log[G : H_S].$$

Modeling requirement

Arbitrary site permutations and exchanges of distinct chemical species are excluded unless they genuinely preserve the background theory.

Symmetry-related states have conjugate stabilizers, so I_G is constant on every G-orbit.

Diagnostics for continuous states and phase limits

Monograph source: Chapter 6, Sections 2–4

diagnostic	records	caution
finite-window rate	thermodynamic scaling of symmetry loss	exhaustion and boundary dependence
quotient G/H and $\dim(G/H)$	continuous broken directions	no statistical weight
$D_{\mathrm{KL}}(\nu \Vert \mu_{G/H})$	concentration relative to invariant phase measure	resolution and measure dependence

Discrete baseline

$\log[G : H]$ remains exact for finite symmetry reductions, point groups, chirality choices, and order–disorder sectors. Continuous symmetry can already occur at finite system size.

Why the diagnostics must remain distinct

Monograph source: Chapter 6, Sections 2 and 4

A broken global $\mathbb{Z}_2$ symmetry has

$$I_L = \log 2, \qquad \frac{I_L}{|X_L|} \longrightarrow 0.$$

The zero rate does not undo the broken discrete symmetry; it says that the orbit choice is subextensive.

For compact G, raw Haar measure of H is positive exactly at finite index. Continuous ordering is instead probed by a phase law ν on G/H:

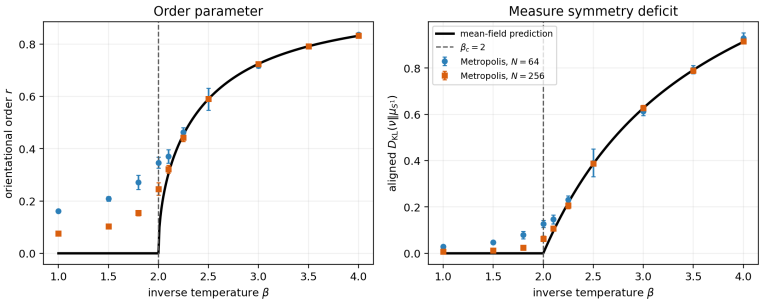
$$I_{\mathrm{meas}}(\nu \mid \mu) = D_{\mathrm{KL}}(\nu \| \mu) \geq 0.$$

Interpretation

Index detects discrete sectors; rate detects scaling; quotient geometry detects broken directions; relative entropy detects statistical selection.

Computed $SO(2)$ orientational transition

Monograph source: Chapter 6, Section 5



$$r = \frac{I_1(\beta J r)}{I_0(\beta J r)}, \qquad \beta_c J = 2.$$

For the selected phase,

$$D_{\mathrm{KL}} = \beta J r^2 - \log I_0(\beta J r).$$

At $\beta = 3$, $N = 256$:

$$r_{\mathrm{sim}} = 0.723995,$$
$$r_{\mathrm{pred}} = 0.724159.$$

Alignment selects a branch and creates the visible finite-N bias below the transition.

Interacting Bose ensembles add a spectral diagnostic

Monograph source: Chapter 6, Section 7

For the one-particle reduced density operator,

$$\gamma^{(1)} = N \operatorname{Tr}_{2,\dots,N} \Gamma, \qquad c_{\mathrm{BEC}} = \liminf \frac{\lambda_{\max}(\gamma^{(1)})}{N}.$$

The cited OpenAI preprint reports (not independently proved here) $c_{\mathrm{BEC}} > 0$ for the exact canonical Gibbs state at a fixed positive density and a positive temperature independent of volume [4].

Shared geometry

A separately phase-selected one-component state has

$$G/H = U(1)/\{1\} \cong S^1,$$

with soft phase twists in its effective Goldstone sector.

Distinct invariants

c_{BEC} is not i_G, and a macroscopic eigenvalue is not the block proxy $(p-q)^2$. A fixed-N canonical state remains $U(1)$-invariant even when it has ODLRO.

One quotient language, two separate theorems

Monograph source: Chapter 6, Section 7

OpenAI result	order statement	main mechanism
interacting hard-sphere Bose gas	macroscopic constant-orbital occupation	thermal paths and density-matrix overlap
quantum Heisenberg ferromagnet	selected zero-field KMS magnetization	exchange paths and boundary pinning

The branch manifolds fit the same reference-group framework:

$$U(1)/\{1\} \cong S^1, \qquad SO(3)/SO(2) \cong S^2.$$

Scope

Shared quotient geometry does not make ODLRO, magnetization, a deficit rate, and a block-density invariant mathematically identical [5].

Roadmap

1 Thesis and scope

2 Finite relational structures

3 Relational ensembles

4 Mean field and critical behavior

5 Continuum and relational geometry

6 Algebra, infinity, and formalization

7 Reference-group phase diagnostics

8 Conclusions

What is established

Monograph source: Synthesis of Chapters 1–6

1. **Finite invariant:** $I(S) = \log(n!/|\operatorname{Aut}(S)|)$ is exactly the logarithm of a relabeling orbit.
2. **Structural calculus:** bounds, rigidity, refinement monotonicity, and preservation results follow from subgroup structure.
3. **Statistical role:** $e^{I(S)}$ is the isomorphism-class multiplicity in relational partition functions.
4. **Explicit transition:** the refined two-block model has an exact antisymmetric instability at $\beta_c = 2/J$ and a positive quartic Landau coefficient.
5. **Extension principle:** operations are functional relations, so the same finite theory applies to algebraic structures.
6. **Reference principle:** physical deficits compare H with $G \leq \operatorname{Aut}(B)$, not with bare-set permutations by default.

The next mathematical problems

Monograph source: Chapter 3, Section 12; Chapters 5–6; Appendix C

- Prove uniform remainder estimates and a genuine local bifurcation theorem for the exact refined free energy.
- Relate microscopic automorphism loss to graphon or kernel variational limits without conflating orbit entropy and block entropy.
- Determine when symmetry-deficit observables have stable finite-size scaling and how their exponents transform under nonlinear proxies.
- Construct a relational renormalization map and identify its fixed points and relevant directions.
- Prove exhaustion-independent limits for finite-window deficits and compare their scaling with conventional susceptibilities.
- Extend formal verification from algebraic derivatives to analytic estimates and limiting arguments.

Symmetry deficit turns the slogan

“structure arises through symmetry breaking”

into an exact finite invariant.

Broader program

Orbit counting gives the finite foundation; variational models show how the invariant can diagnose organized phases. Physical applications specify a structured reference group already at finite size, then use index, rate, geometric, or measure diagnostics as the state space and limits require.

Selected references I

[1] Paul Erdős and Alfréd Rényi.
Asymmetric graphs.
Acta Mathematica Academiae Scientiarum Hungaricae, 14:295–315, 1963.
https://doi.org/10.1007/BF01895716.

[2] Jeffrey Goldstone, Abdus Salam, and Steven Weinberg.
Broken symmetries.
Physical Review, 127(3):965–970, 1962.
https://doi.org/10.1103/PhysRev.127.965.

[3] László Lovász and Balázs Szegedy.
Limits of dense graph sequences.
Journal of Combinatorial Theory, Series B, 96(6):933–957, 2006.
https://doi.org/10.1016/j.jctb.2006.05.002.

[4] OpenAI.
Bose–Einstein condensation at positive temperature in the dilute hard-sphere gas, 2026.
OpenAI Math Release preprint, October 5, 2026. Online manuscript.

[5] OpenAI.
Spontaneous magnetization in the quantum Heisenberg ferromagnet, 2026.
OpenAI Math Release preprint, September 24, 2026. Online manuscript.

Selected references II

[6] Chen Ning Yang and Robert L. Mills.
Conservation of isotopic spin and isotopic gauge invariance.
Physical Review, 96(1):191–195, 1954.
https://doi.org/10.1103/PhysRev.96.191.

Roadmap

9 Backup

Backup: exact derivatives on the contrast slice

Monograph source: Chapter 3, Section 5

For

$$h(u) = -u \log u - (1-u)\log(1-u),$$

$$h''(u) = -\frac{1}{u(1-u)}, \qquad h^{(4)}(u) = -\frac{2(1-3u+3u^2)}{u^3(1-u)^3}.$$

At $u = 1/2$,

$$h''(1/2) = -4, \qquad h^{(4)}(1/2) = -32.$$

Hence

$$\mathcal{G}''_{\beta,J}(0) = 2(4 - 2\beta J), \qquad \mathcal{G}^{(4)}_{\beta,J}(0) = 64 = 24\left(\frac{8}{3}\right).$$

Backup: representative numerical minimizers

Monograph source: Chapter 3, Section 11; Appendix B

β	p_*	q_*	$\Delta = \lvert p_* - q_* \rvert$	phase
1.500	0.500000	0.500000	0	symmetric
2.000	0.500000	0.500000	0	threshold grid point
2.025	0.595760	0.404240	0.191520	broken
2.500	0.855205	0.144795	0.710411	broken
3.000	0.929280	0.070720	0.858560	broken

The symmetry-related companion minimizer is obtained by exchanging p_* and q_*. The grid establishes agreement to its resolution, not an independent numerical proof of the exact threshold.

Backup: exact finite compatibility

Monograph source: Chapter 5, Section 3

Suppose an infinite theory assigns a deficit $\mathcal{I}(G, H)$ to residual symmetry $H \leq G$. At minimum, finite compatibility asks for

$$\mathcal{I}(G, H) = \log[G : H] \qquad \text{when } [G : H] < \infty.$$

In the original setting,

$$G = \mathrm{Sym}(X), \qquad H = \mathrm{Aut}(S),$$

so

$$\mathcal{I}(G, H) = \log \frac{|G|}{|H|} = I(S).$$

Dimension, quotient measure, and entropy-rate proposals should therefore be viewed as category-specific continuations of the orbit-size principle.

Backup: notation distinguishes three diagnostics

Monograph source: Chapters 1 and 3; Appendices

symbol here	definition	role
$I(S)$	$\log(n!/\lvert\operatorname{Aut}(S)\rvert)$	exact finite invariant
I_Γ	$\log(\lvert\Gamma\rvert/\lvert\operatorname{Stab}_\Gamma\rvert)$	reduced discrete invariant
D_{cont}	$(p-q)^2$	continuous two-block proxy
D_{lat}	$q^{-1}\sum_i (f_i - q^{-1})^2$	lattice proxy

These quantities agree on which simple phases are symmetric, but they are not numerically identical and need not share critical exponents.